Critical Replication: SimpleTM for Multivariate Time Series Forecasting

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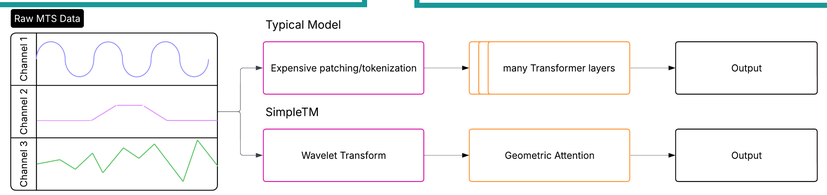
Intro: Transformers in Multivariate Time-Series Forecasting

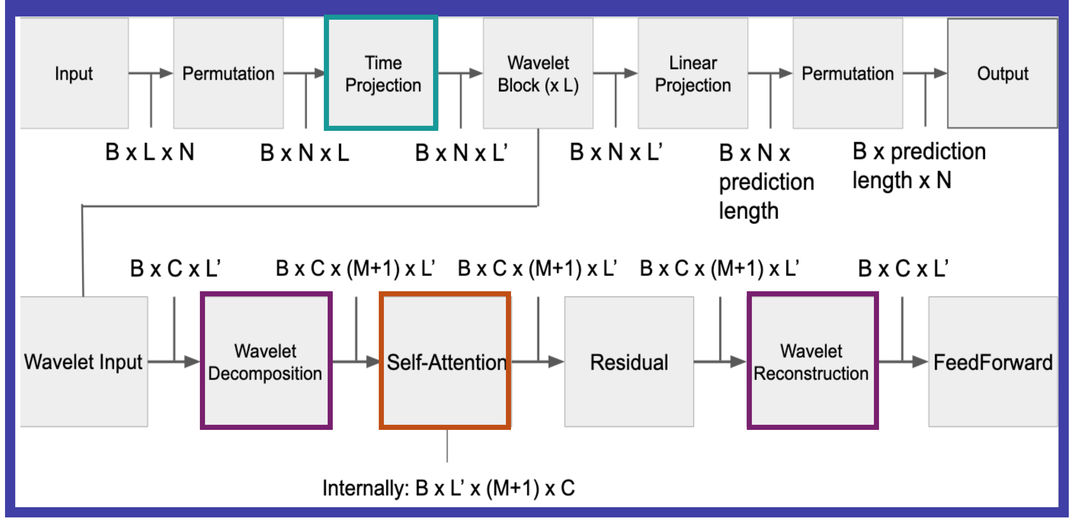
Self-attention in transformer models facilitate mixing of temporal information in time-series forecasting

iTransformer: variate-wise attention TimeMixer: multi-scale decomposition and mixing



Is it possible to simplify the job of the model with clever, informative inductive biases?

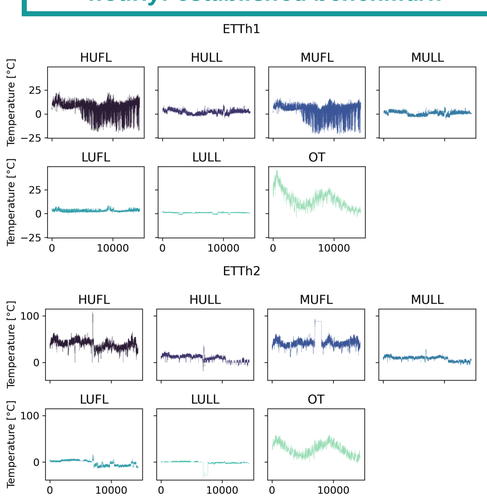
SimpleTM: Multivariate Time-Series Forecasting Featuring Wavelet Transform and Geometric Attention

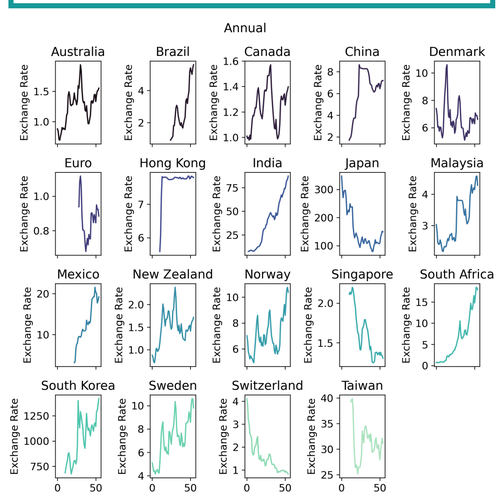


Datasets and Baseline Results

electricity transformer temperature (ETT) hourly: established benchmark

annual national currency exchange rates: new challenge (limited lookback window with significant variability)





Dataset	Baseline MSE	Baseline MAE	Author MSE	Author MAE
ETTh1	0.3877±0.0030	0.4061±0.0023	0.366	0.392
ETTh2	0.2956±0.0022	0.3476±0.0019	0.281	0.338

Evaluating the Weight of Geometric Attention on ETTh1/2 and Exchange Rate Datasets

geometric attention: simultaneously captures similarity and complementarity across channels through the geometric product

alpha: modulates relative weight of dot and wedge product

the geometric attention mechanism does not substantially impact error on ETTh1/2 and exchange rate data

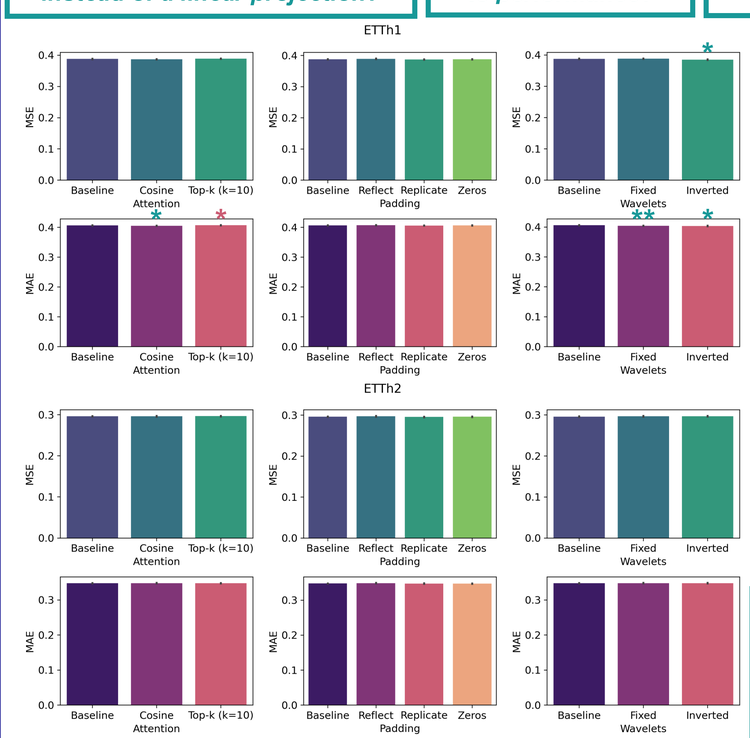
Wavelet and Attention-Based Ablations On ETTh1/2 Datasets

Can the wavelet transform be applied to the original time series instead of a linear projection?

Are learnable wavelet parameters essential to performance?

Do attention mechanisms capturing alternative token complementarities impact model performance?

Is the assumption of signal periodicity harmful?



Dataset	Ablation Type	Ablation	MSE	MAE	MSE p-value (paired t-test)	MAE p-value (paired t-test)
ETTh1	Attention	Cosine	0.3869±0.0023	0.4051±0.0019	0.1732	0.0191
		Top-k (k=10)	0.3886±0.0028	0.4072±0.0021	0.1216	0.0065
	Padding	Reflect	0.3893±0.0039	0.4069±0.0025	0.1482	0.2570
		Replicate	0.3870±0.0028	0.4058±0.0018	0.3347	0.6575
		Zeros	0.3879±0.0024	0.4061±0.0020	0.8741	0.9731
	Wavelets	Fixed	0.3882±0.0022	0.4041±0.0017	0.6043	0.0030
		Inverted	0.3851±0.0043	0.4035±0.0035	0.0467	0.0334
ETTh2	Attention	Cosine	0.2962±0.0036	0.3479±0.0026	0.8072	0.6668
		Top-k (k=10)	0.2964±0.0022	0.3479±0.0022	0.4601	0.5375
	Padding	Reflect	0.2972±0.0039	0.3486±0.0030	0.1382	0.1438
		Replicate	0.2955±0.0031	0.3475±0.0024	0.4253	0.9078
		Zeros	0.2960±0.0034	0.3474±0.0026	0.9618	0.7518
	Wavelets	Fixed	0.2968±0.0035	0.3478±0.0026	0.3506	0.7287
		Inverted	0.2969±0.0034	0.3478±0.0027	0.4116	0.8341

alternate, non-geometric attention mechanisms, fixed wavelet parameters, and application of wavelet transform on original time signal → small but statistically meaningful changes in squared/absolute error for ETTh1

Conclusion and Future Directions

The SimpleTM architecture is generally robust, while the importance of specific architectural details likely vary according to dataset hyperparameter synergy

Can we go simpler? → ARIMA, MLP, and naïve benchmarks

Mind the gap → investigate the discrepancy in baseline/author MSE/MAEs

Is it wavelet, or is it convolution with dilation? → stacked convolutions without recursive decomposition/wavelet initialization

References